The Quadratic Formula

Name _____

Algebra 2 Lesson 2-6 (fast track) Date _____ Period _____

Learning goal: I can solve quadratic equations using the Quadratic Formula and use the discriminant to predict the number and type of solutions.

Language goal: I can explain how many solutions a quadratic equation has and what type they are, using the words discriminant, radicand, real, and non-real.

On the quiz: (1) Solve with the Quadratic Formula; pick the simplified solutions (Example 1A). (2) Bonus: the same with a negative radicand, so the solutions have i (Example 1B). (3) Use the discriminant to tell how many roots and what type (Example 3). (4) Will a tossed ball reach a height? (full Savvas quiz only) (5) Which b make the discriminant 0? (full Savvas quiz only)

The Quadratic Formula. For any quadratic equation $ax^2 + bx + c = 0$ with $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression under the radical, $b^2 - 4ac$, is the **radicand**. It is also called the **discriminant**.

Example 1: Solve Quadratic Equations

What are the solutions to the equation? Write the equation as $ax^2 + bx + c = 0$, name a, b, c, then substitute. Use parentheses around negative numbers.

A. $3x^2 - 4x - 9 = 0$
$a = 3,\ \ b = -4,\ \ c = -9$

$$\begin{aligned} x &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-9)}}{2(3)} \\ &= \frac{4 \pm \sqrt{16 + 108}}{6} = \frac{4 \pm \sqrt{124}}{6} \\ &= \frac{4 \pm 2\sqrt{31}}{6} \qquad (\sqrt{124} = \sqrt{4 \cdot 31}) \\ &= \frac{2 \pm \sqrt{31}}{3} \end{aligned}$$

The radicand is $124 > 0$: **two real** solutions,

$$x = \frac{2 + \sqrt{31}}{3} \quad \text{and} \quad x = \frac{2 - \sqrt{31}}{3}.$$

B. $x^2 - 9x + 27 = 0$
$a = 1,\ \ b = -9,\ \ c = 27$

$$\begin{aligned} x &= \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(27)}}{2(1)} \\ &= \frac{9 \pm \sqrt{81 - 108}}{2} = \frac{9 \pm \sqrt{-27}}{2} \\ &= \frac{9 \pm i\sqrt{27}}{2} \qquad (\sqrt{-1} = i) \\ &= \frac{9 \pm 3i\sqrt{3}}{2} \qquad (\sqrt{27} = 3\sqrt{3}) \end{aligned}$$

The radicand is $-27 < 0$: **two non-real** solutions,

$$x = \frac{9 + 3i\sqrt{3}}{2} \quad \text{and} \quad x = \frac{9 - 3i\sqrt{3}}{2}.$$

Say it: In Part B the radicand is _____. It is _____ (positive / negative), so the solutions are _____ (real / non-real) because ...

Try It 1: Solve using the Quadratic Formula

a. $2x^2 + 6x + 3 = 0$

$a =$ _____ $b =$ _____ $c =$ _____

$$x = \frac{-(\quad) \pm \sqrt{(\quad)^2 - 4(\quad)(\quad)}}{2(\quad)}$$

Radicand: _____

Real or non-real? _____

$x =$ _____

b. $3x^2 - 2x + 7 = 0$

$a =$ _____ $b =$ _____ $c =$ _____

$$x = \frac{-(\quad) \pm \sqrt{(\quad)^2 - 4(\quad)(\quad)}}{2(\quad)}$$

Radicand: _____

Real or non-real? _____

$x =$ _____

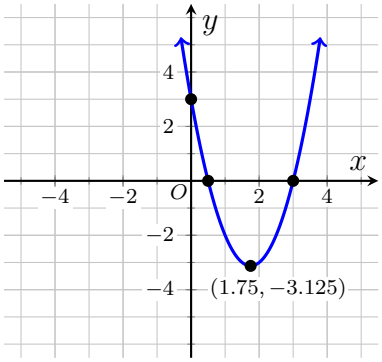
In part b, where does the i come from? Use the word *radicand*.

Challenge (DOK 3, self-paced). Seaira wants to use the Quadratic Formula to solve the equation $x^4 + 5x^2 - 5 = 0$. Is this possible? If so, describe the steps she should follow.

Example 3: Identify the Number of Real-Number Solutions

How can you determine the number and type of roots for a quadratic equation? Graph each equation. Then use the Quadratic Formula to find the roots.

$y = 2x^2 - 7x + 3$



$2x^2 - 7x + 3 = 0$

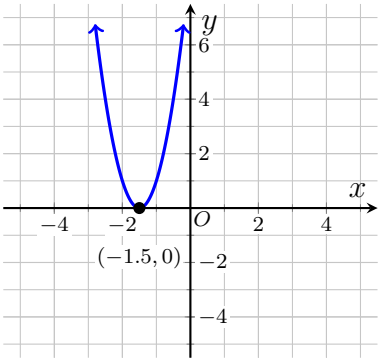
$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(2)(3)}}{2(2)}$$

$$x = \frac{7 \pm \sqrt{25}}{4} = \frac{7 \pm 5}{4}$$

$x = 3$ and $x = \frac{1}{2}$

The radicand is 25. Adding and subtracting $\sqrt{25}$ gives two different roots. The graph shows **two** x-intercepts: **two real** roots.

$y = 4x^2 + 12x + 9$



$4x^2 + 12x + 9 = 0$

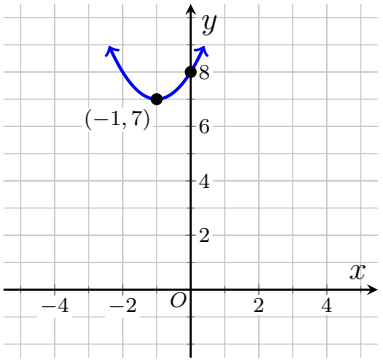
$$x = \frac{-12 \pm \sqrt{(12)^2 - 4(4)(9)}}{2(4)}$$

$$x = \frac{-12 \pm \sqrt{0}}{8} = \frac{-12}{8}$$

$x = -\frac{3}{2}$

The radicand is 0. Adding and subtracting $\sqrt{0}$ gives one root. The graph shows **one** x-intercept (the vertex): **one real** root.

$y = x^2 + 2x + 8$



$x^2 + 2x + 8 = 0$

$$x = \frac{-2 \pm \sqrt{(2)^2 - 4(1)(8)}}{2(1)}$$

$$x = \frac{-2 \pm \sqrt{-28}}{2} = \frac{-2 \pm 2i\sqrt{7}}{2}$$

$x = -1 \pm i\sqrt{7}$

The radicand is -28. Its square root is imaginary. The graph shows **no** x-intercepts: **two non-real** roots.

The **discriminant** is the radicand, $b^2 - 4ac$ (a, b, c real, $a \neq 0$). It tells the nature of the roots:
$b^2 - 4ac > 0$: two real roots $b^2 - 4ac = 0$: one real root $b^2 - 4ac < 0$: two non-real roots

Say it: For $x^2 + 2x + 8 = 0$ the discriminant is _____. It is _____, so the equation has _____ _____ roots, and the graph _____ the x-axis.

Try It 3: Describe the nature of the solutions for each equation

a. $16x^2 + 8x + 1 = 0$
$a =$ _____ $b =$ _____ $c =$ _____
$b^2 - 4ac =$

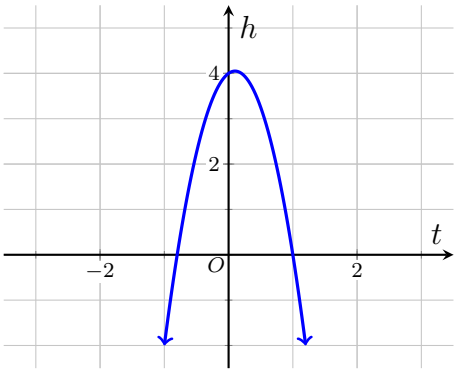
The discriminant is _____ (> 0, $= 0$, < 0), so there is/are _____ solution(s).

b. $2x^2 - 5x + 6 = 0$
$a =$ _____ $b =$ _____ $c =$ _____
$b^2 - 4ac =$

The discriminant is _____ (> 0, $= 0$, < 0), so there is/are _____ solution(s).

Practice: Will the ball reach that height?

Deon throws a ball into the air. The height, h, of the ball, in meters, at time t seconds is modeled by the function $h(t) = -5t^2 + t + 4$.



a. When will the ball hit the ground? (Hit the ground: $h = 0$.)

b. Will the ball reach a height of 5 meters?
Set $h(t) = 5$: _____
Standard form: _____ $= 0$
$a =$ ____ $b =$ ____ $c =$ ____ $b^2 - 4ac =$ _____
So:

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
YQH ★	Solve a quadratic equation using the quadratic formula	Quiz 1; Quiz 2 (bonus). Example 1, Try It 1
QHK ★	Using the discriminant	Quiz 3. Example 3, Try It 3 (also Quiz 4, 5: full Savvas quiz only)
5VV	Introduction to complex numbers	Optional warm-up, no quiz score: square roots of negatives (Example 1B)
CEU	Solve quadratic equations: word problems	Optional: Practice page; Quiz 4 (full Savvas quiz only)

Prerequisites if you need review (no quiz score): CJC solve by factoring, NPH completing the square.

Before the quiz: Finish these sentences

To use the Quadratic Formula, first write the equation in the form _____ $= 0$.

The radicand of the Quadratic Formula, $b^2 - 4ac$, is called the _____.

If the discriminant is positive, the equation has _____. If it is zero, it has _____.

If the discriminant is negative, the equation has two _____ solutions, and the graph does not _____ the x-axis.